# Hybrid Waiting Lists

A viral new Thai noodle restaurant has a huge number of customers piling up in front of their door. The restaurant implemented two systems for customer to be put in the Waiting List: Online Reservation and Walk-In. Usually, the customer who have booked online will be prioritized, however for balancing purpose as the restaurant receive huge load of both online and walk-in, the newly opened restaurant decides to serve the customers alternately between the 2 waiting lists with **First Come First Serve** basis. Because the long waiting time (2-3 hours), there are also customers who choose to cancel.

## Problem Description

Given the number of Online Reservation customers $N$, Walk-In customers $M$, and cancelling customers $L$ you will be asked for a unique integer id for each online customer $(o_1, o_2, ..., o_N)$, walk-in customer $(w_1, w_2, ..., w_M)$, and cancelling customer $(c_1, c_2, ..., c_L)$. The task is to list out the serving order until all the customers are served.

## Input Format

- The first line consists of two integers $N$ and $M$, where $N$ is an integer for number of Online Reservation Customers and $M$ is an integer for number of Walk-In Customers.
- The second line consists of $N$ values representing the unique ids of Online Reservation customers.
- The third line consists of $M$ values representing the unique ids of Walk-In customer.
- The fourth line consists of an integer $L$, representing the number of customer who cancelled.
- Lastly, the fifth line consists of $L$ values representing the unique ids of customer who cancelled.

## Output Format

List out the id of the customers in the order of serving until all customers are served, starting from the Online Reservation customer. The customer who cancelled will be gone from this list.

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## Constraints

- $1 \leq N, M \leq 1000000$
- $0 \leq L \leq N + M - 1$
- Implement your algorithm in C using Single Linked List (SLL) insertion and deletion.
- There should be 3 SLL, representing 3 queues: Online, Walk-In and Hybrid.

## Sample Input 1 (standard input)

```
2 2
10 15
20 13
0
```

## Sample Output 1 (standard output)

```
10 20 15 13
```

## Sample Input 2 (standard input)

```
6 3
10 8 9 20 19 16
34 27 11
3
10 16 11
```

## Sample Output 2 (standard output)

```
34 8 27 9 20 19
```

## Explanation 2

- First, we have 6 Online customers and 3 Walk-in customers.
- 10, 8, 9, 20, 19 and 16 are inputted as the id of online customers, then 34, 27 and 11 are inputted as the id of walk-in customers.
- With both of those id, we can set the order alternately starting from online customer which resulted in 10, 34, 8, 27, 9, 11, 20, 19, and 16. Because there are no more walk-in customer after 20, customer id 19 and 16 can be served.

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- However, there are 3 customers which cancelled with id 10, 16 and 11, so we delete the id from the previous order to become 34, 8, 27, 9, 20, and 19.

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